Using PageRank to Identify Critical Airports

Applied Linear Algebra course project

Airport route network • PageRank • Hub disruption

549 airports

5,450 directed routes

ATL closure scenario

Source: OpenFlights; project notebook

Team: Mindeok Seo, Jake Gust, Jeewan Khadka, Yijia Zhang, Alan Tang

PageRank converts a route map into a ranking of structural importance.

Instead of only counting routes, it asks whether an airport is connected to other important airports.

Research Question

Why PageRank is useful for transportation networks

Which airports are most important when importance is measured by network structure, not just direct route count?

Traditional view

Rank airports by number of direct routes. This measures connectivity but treats all connected airports equally.

PageRank view

Rank airports by recursive importance. Connections from important airports count more than isolated connections.

Project extension

Remove a major hub and recompute PageRank to see how importance redistributes across the network.

Core idea: an airport is central if it receives connections from airports that are themselves central.

Source: OpenFlights; project notebook

Data and Network Setup

OpenFlights route data filtered to direct routes between U.S. airports

549
airports in network

5,450
directed route edges

549×549
adjacency matrix

7
dangling airports

Binary model

Route = 1
No route = 0

Filtering decisions

Only U.S. airports with valid IATA codes are included. Routes are kept only when both source and destination are U.S. airports, and the route has zero stops.

Interpretation

Because the matrix is binary, the model measures route connectivity rather than passenger volume, aircraft capacity, or flight frequency.

Source: OpenFlights; project notebook

From Routes to Matrices

The network becomes a linear algebra problem

1		2		3		4
Directed graph	→	**Adjacency matrix**	→	**Transition matrix**	→	**PageRank vector**
Airports are nodes; routes are directed edges.		A[i,j] = 1 if route j → i exists.		Normalize columns so sum_i M[i,j] = 1.		Power iteration finds the steady-state vector.

At convergence: Gv = v The final vector is the principal eigenvector of the Google Matrix.

Source: OpenFlights; project notebook

PageRank Method

Power iteration with damping

Update equation

$$v_{k+1} = dM\,v_k + ((1 - d)/N)e$$

d = 0.85 damping factor
85% follows actual routes
15% randomly jumps to any airport
Stop when L1 change is below tolerance

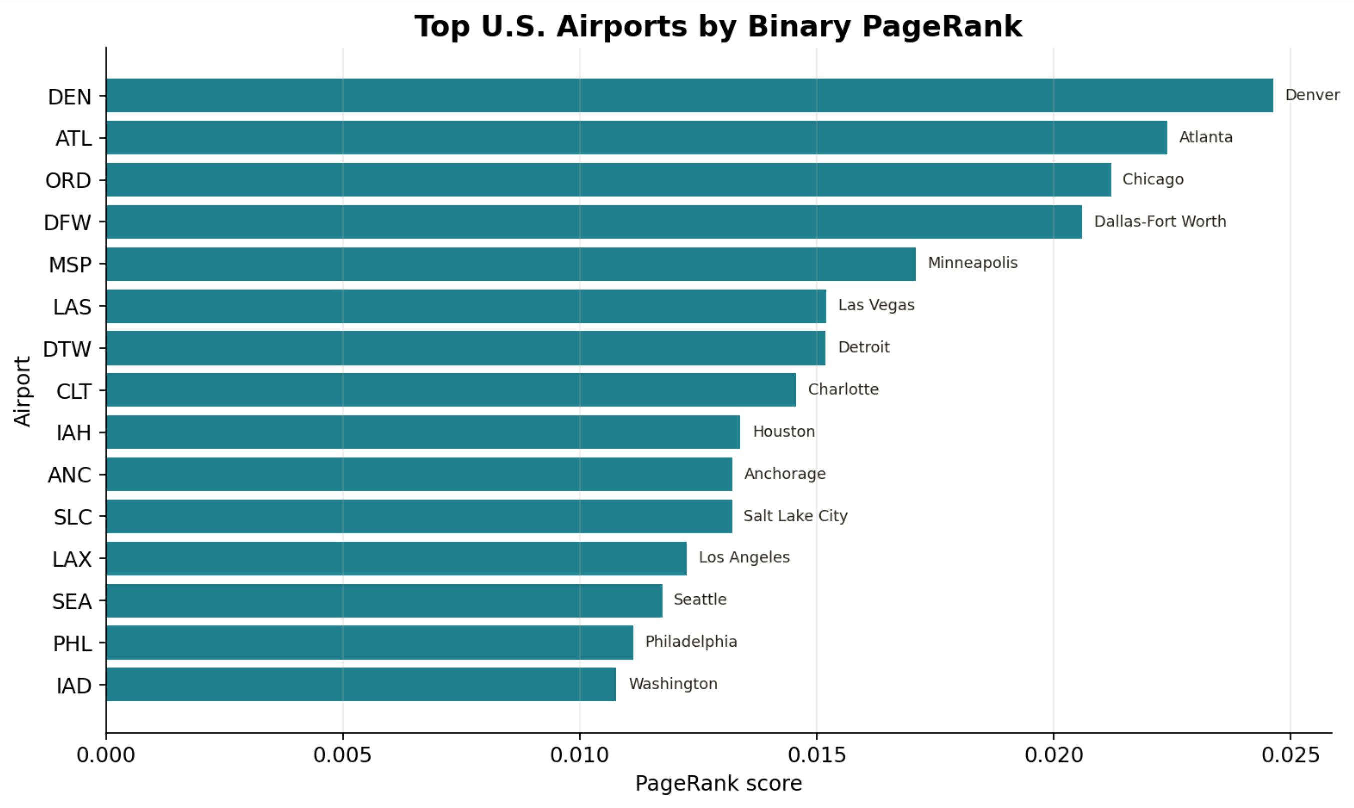
Why damping matters

Prevents dead ends from trapping the model
Makes the Markov chain more stable
Ensures the vector converges
Allows every airport to retain some probability

Result: one PageRank score for every airport.

Baseline Results

DEN ranks first in the binary route network



Top PageRank airports

DEN has the highest PageRank score.
ATL has the highest total degree, but ranks second.
ORD and DFW follow closely.
PageRank is similar to degree centrality but not identical.

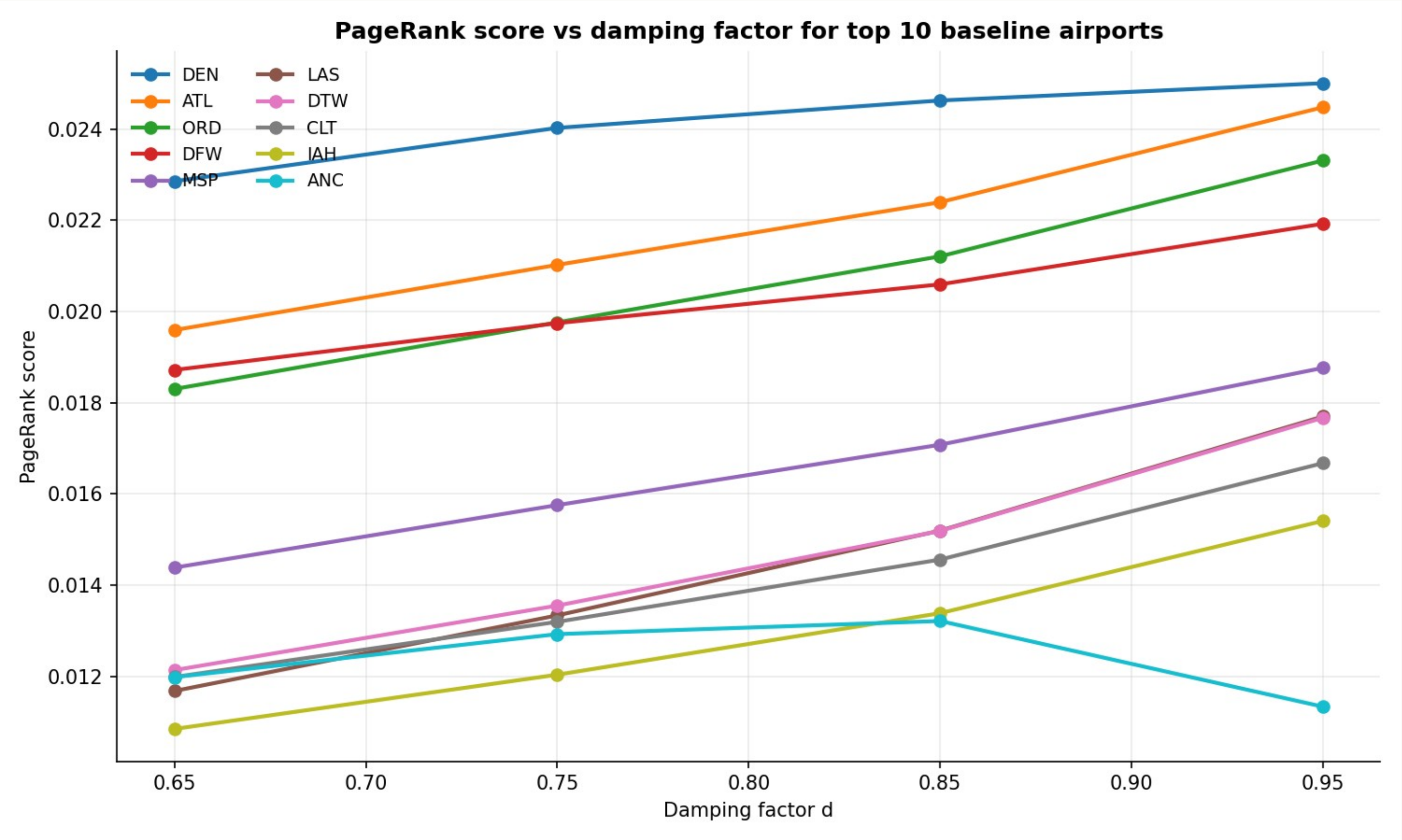
Interpretation

DEN's centrality reflects strong connections across important parts of the U.S. route network.

Source: OpenFlights; project notebook

Damping Factor Sensitivity

Are the top hubs stable across choices of d?



Setup

Recompute PageRank at d = 0.65, 0.75, 0.85, 0.95.
Same transition matrix, same tolerance.
Compare top hubs and full ranking.

Findings

DEN, ATL, ORD, DFW stay top 4 for every d.
Spearman ρ ≥ 0.988 vs baseline.
Top-10 set overlaps 9 of 10 in every case.
Iterations rise: 43 → 64 → 112 → 353.

Top hubs are driven by network structure, not by d = 0.85.

Source: OpenFlights; project notebook

Hub Closure Experiment

What happens when ATL is removed?

Simulation: remove all incoming and outgoing ATL routes, rebuild the matrices, and recompute PageRank.

ATL	5,145	112	DEN
removed hub	remaining directed routes	post-closure iterations	still ranked #1

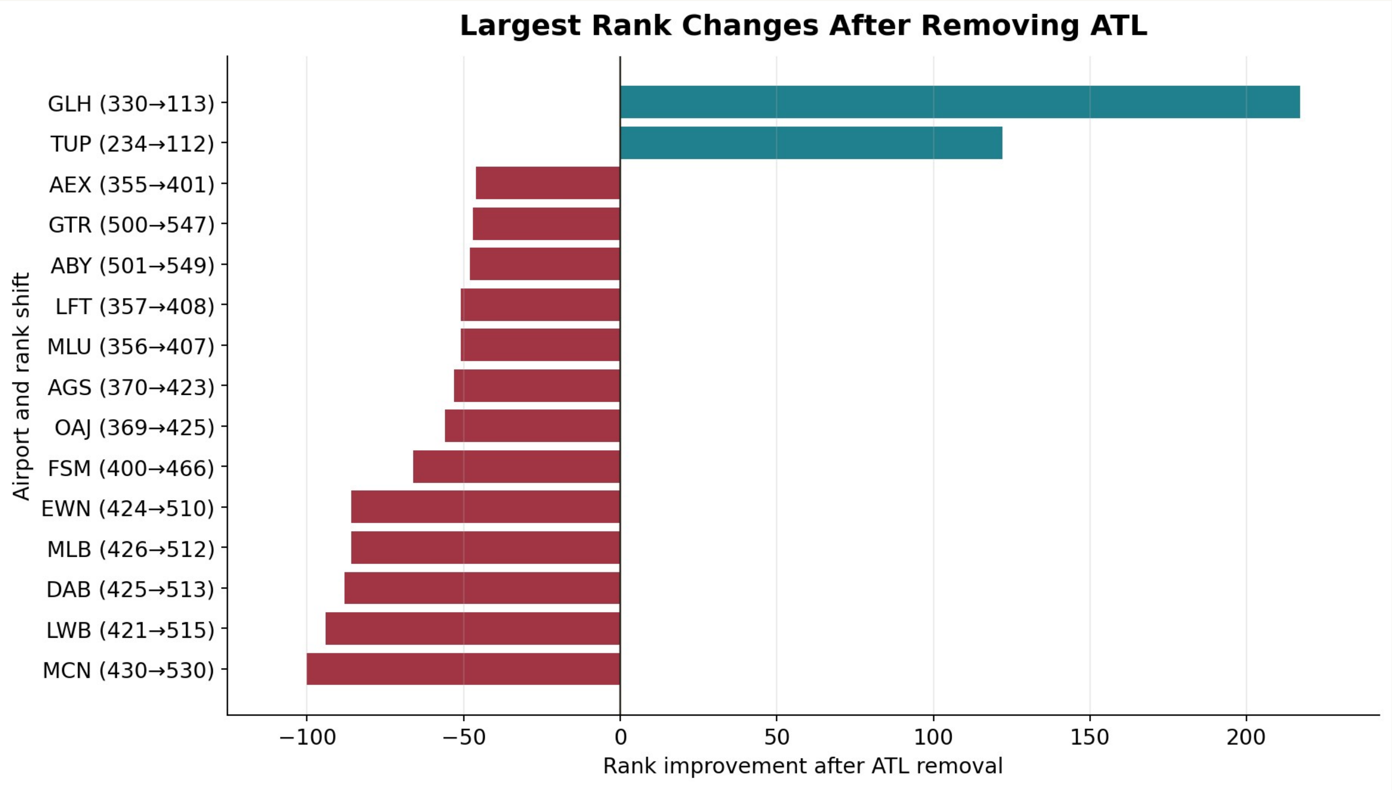
Main finding

The network does not collapse. Importance redistributes toward other major hubs, especially DEN, ORD, DFW, MSP, DTW, CLT, and IAH.

Source: OpenFlights; project notebook

Rank Changes After Removing ATL

Largest movements in the disrupted network



How to read this

Positive rank change means the airport moved up.
Negative rank change means it moved down.
Small airports can jump many ranks because bottom scores are very close.

Key message

For hub-level interpretation, PageRank score change matters more than rank alone.

What We Learned

Centrality and resilience in one model

1. Centrality is recursive

An airport is important when it is connected to other important airports, not only when it has many direct routes.

2. Robust to damping

Top hubs stay the same for d = 0.65 through 0.95. The ranking reflects network structure, not the choice of d.

3. The network is resilient

Removing ATL changes rankings, but other hubs absorb importance. The national structure stays connected through multiple hubs.

4. Local vulnerability remains

Some regional airports lose importance when ATL is removed, showing that local dependence can be high even if the full network is resilient.

PageRank gives a linear algebra-based way to quantify both structural importance and disruption response.

Source: OpenFlights; project notebook

Limitations and Extensions

How the project could be strengthened

Limitations

OpenFlights route data is historical.
Binary matrix does not measure passenger volume.
All outgoing routes are treated as equally likely.
The ATL scenario is a complete closure, not a partial disruption.

Extensions

Use BTS T-100 data to weight edges by passengers or flight counts.
Compare PageRank with degree, betweenness, and eigenvector centrality.
Simulate multiple hub closures.
Map centrality shifts geographically.

Source: OpenFlights; project notebook

Conclusion

Airport importance is not just route count. It depends on where an airport sits inside the network of important connections.

PageRank turns that idea into a matrix problem: construct the transition matrix, apply damping, iterate, and interpret the principal eigenvector.

Final takeaway

DEN, ATL, ORD, and DFW are structurally central. Removing ATL shifts importance to other hubs, showing both centrality and resilience.

Team: Mindeok Seo, Jake Gust, Jeewan Khadka, Yijia Zhang, Alan Tang

Repo: github.com/JeewaanK/Airport_PageRank